# JARED COBRIN

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## EDUCATION

**Concordia University** — Montreal, Quebec
*BCompSc in Computer Science* — *Jan 2025 – present*

- GPA: 3.28/4.30
- Relevant courses: Deep Learning, Machine Learning, Multivariable Calculus I, Applied Linear Algebra

**Athabasca University** — Online
*Commerce* — *Sept 2024 – Dec 2024*

- GPA: 3.85/4.00

**John Abbott College** — Montreal, Quebec
*Commerce* — *Aug 2021 – May 2022*

- R-Score: 31.6

## RESEARCH

**Causal Natural Language Autoencoders for Collusion Monitoring**
*Apart Research AI Collusion Research Sprint* — *Oct 2026*

- Built from scratch an activation verbalizer grounded in dozens of verified probe directions, trained with a custom four-part reinforcement learning reward that never names a concept.
- It names concepts it was never taught and flags collusive sentences 14.3/20 vs 3.0/20 for honest ones (Qwen2.5-7B-Instruct); paper and code public.

**SAE-Grounded Evaluation of NLA Explanations**
*Independent research* — *Sep 2026*

- Used sparse autoencoders to test whether NLA explanations reflect what is in the activation across 200 explanations: the first two paragraphs recover 30% of latents vs 70% for the full text.
- Showed reconstruction score does not predict groundedness ($r = -0.01$, $n = 188$); write-up and code public.

**Choosing the Number of Principal Components in Optimal-Transport Steering**
*Extension of Nanfack et al. (2026)* — *Sep 2026*

- Reproduced PCA-OT refusal ablation on Llama-2-13B and Qwen2.5-14B; doubling the training set did not clearly help, as draws varied as much as the doubling.
- Extended it to sycophancy on Llama-2-7B: PCA-OT only steers at K=20-50, because the signal is spread across many PCA directions (99.9% in the first direction for refusal vs 2.7% for sycophancy).

## EXPERIENCE

**Spotless Shine** — Montreal, Quebec
*Founder / Software Engineer* — *May – Aug 2025*, *May – Aug 2026*

- Founded and operated a business with 5-figure monthly revenue, managing a team of over 10 employees and leading door-to-door sales.
- Engineered the company's full-stack platform (React, TypeScript, FastAPI, PostgreSQL) with three user roles (owner, door knocker, cleaner), a lead-to-paid job pipeline, scheduling and commission tracking.
- Built a custom canvassing map with lasso selection, territory assignment and live multi-user sync, replacing paid subscriptions and saving hundreds of dollars a month.

**Allin1Gaming** — Montreal, Quebec
*Product Design Intern* — *Apr 2024 – Apr 2025*

- Designed a modular magnetic gaming desk system through materials research, cost analysis and first-principles analysis; conducted market and competitor analysis.

## COURSES

- **ARENA:** Interpretability
- **Andrej Karpathy:** Neural Networks Zero to Hero
- **Coursera:** Machine Learning Specialization

## LEADERSHIP AND ACTIVITIES

**ENACTUS Concordia** Montreal, Quebec
*VP, SustainaBite* *May 2025 – Dec 2025*

- Led SustainaBite, a social-impact venture focused on reducing campus waste and food insecurity.

**Junior Hockey, EHL (Pennsylvania Huntsmen)** Pennsylvania & New Jersey
*Player* *Jun 2022 – Mar 2024*

- Developed a strong work ethic through a rigorous personal training regimen alongside mandatory team practices.

## SKILLS

**Programming:** Python, Java, JavaScript

**ML / AI:** PyTorch, Hugging Face Transformers, LoRA/PEFT, SGLang, scikit-learn, NumPy, Sparse Autoencoders, Linear Probes, Activation Steering, Reinforcement Learning, LLM Evaluation

**Development:** Git/GitHub, React, Firebase, FastAPI, PostgreSQL